

Ex. No.: 5

SELECT with various clause – WHERE, pattern matching

```
mysql> SELECT RegNo, Name, DOB
-> FROM Student;
```

RegNo	Name	DOB
191711341	Ramesh	2002-04-15
191711342	Harish	2001-08-20
191711343	Aish	2002-01-25
191711344	Priya	2002-06-18
191711345	Shyam	2001-12-05
191711346	Kamal	2002-02-28
191711347	Ram	2001-10-10

```
7 rows in set (0.004 sec)
```

```
mysql> SELECT *
-> FROM Student
-> WHERE Gender='Female';
```

RegNo	Name	Gender	DOB	City	CourseID
191711343	Aish	Female	2002-01-25	Salem	C001
191711344	Priya	Female	2002-06-18	Coimbatore	C003

```
2 rows in set (0.006 sec)
```

```
mysql> SELECT *
-> FROM Student
-> WHERE CourseID='C001';
```

RegNo	Name	Gender	DOB	City	CourseID
191711341	Ramesh	Male	2002-04-15	Salem	C001
191711343	Aish	Female	2002-01-25	Salem	C001
191711345	Shyam	Male	2001-12-05	Salem	C001

```
3 rows in set (0.007 sec)
```

```
mysql> SELECT *
-> FROM Faculty
-> WHERE DOJ < '2014-11-01';
```

FacultyID	Name	DOJ
101	Mohan	2012-05-20
103	Ravi	2014-10-10
104	Kumar	2013-12-01

```
rows in set (0.007 sec)
```

```
mysql> SELECT *
-> FROM Course
-> WHERE HallID<>'H001';
```

CourseID	CourseName	HallID
C002	Java	H002
C003	DBMS	H003
C004	AI	H002

3 rows in set (0.008 sec)

```
mysql> SELECT *
-> FROM Student
-> WHERE Name LIKE '%sh';
```

RegNo	Name	Gender	DOB	City	CourseID
191711341	Ramesh	Male	2002-04-15	Salem	C001
191711342	Harish	Male	2001-08-20	Chennai	C002
191711343	Aish	Female	2002-01-25	Salem	C001

3 rows in set (0.052 sec)

```
mysql> SELECT *
-> FROM Student
-> WHERE Name LIKE '%sh%';
```

RegNo	Name	Gender	DOB	City	CourseID
191711341	Ramesh	Male	2002-04-15	Salem	C001
191711342	Harish	Male	2001-08-20	Chennai	C002
191711343	Aish	Female	2002-01-25	Salem	C001
191711345	Shyam	Male	2001-12-05	Salem	C001

4 rows in set (0.005 sec)

```
mysql> SELECT *
-> FROM Student
-> WHERE City LIKE '%Sal%';
```

RegNo	Name	Gender	DOB	City	CourseID
191711341	Ramesh	Male	2002-04-15	Salem	C001
191711343	Aish	Female	2002-01-25	Salem	C001
191711345	Shyam	Male	2001-12-05	Salem	C001
191711347	Ram	Male	2001-10-10	Salem	C003

4 rows in set (0.006 sec)

```
mysql> SELECT *
-> FROM Student
-> WHERE Name NOT LIKE '_____';
```

RegNo	Name	Gender	DOB	City	CourseID
191711343	Aish	Female	2002-01-25	Salem	C001
191711344	Priya	Female	2002-06-18	Coimbatore	C003
191711345	Shyam	Male	2001-12-05	Salem	C001
191711346	Kamal	Male	2002-02-28	Madurai	C002
191711347	Ram	Male	2001-10-10	Salem	C003

5 rows in set (0.006 sec)

```
mysql> SELECT *
-> FROM Student
-> WHERE Name LIKE '%am%';
```

RegNo	Name	Gender	DOB	City	CourseID
191711341	Ramesh	Male	2002-04-15	Salem	C001
191711345	Shyam	Male	2001-12-05	Salem	C001
191711346	Kamal	Male	2002-02-28	Madurai	C002
191711347	Ram	Male	2001-10-10	Salem	C003

4 rows in set (0.009 sec)

EXP.NO:6 SELECT with various clause – BETWEEN, IN, Aggregate function

1)

```
mysql> SELECT CourseName, CourseType
-> FROM Course
-> WHERE CourseName IN ('Statistics','Programming');
```

CourseName	CourseType
Statistics	Theory
Programming	Practical

2 rows in set (0.005 sec)

2)

```
mysql> SELECT CourseNo, Score
-> FROM StudentMarks
-> WHERE Score BETWEEN 50 AND 80;
```

CourseNo	Score
C001	75
C002	65
C001	55
C002	72
C003	67
C002	79
C001	75
C002	65
C001	55
C002	72
C003	67
C002	79

```
12 rows in set (0.009 sec)
```

3)

```
mysql> SELECT AVG(Score) AS AverageMark
-> FROM StudentMarks
-> WHERE CourseNo='C002';
```

AverageMark
69.2000

```
1 row in set (0.019 sec)
```

4)

```
mysql> SELECT MAX(Score) AS MaximumMark,
-> MIN(Score) AS MinimumMark
-> FROM StudentMarks
-> WHERE CourseNo='C002';
```

MaximumMark	MinimumMark
82	48

```
1 row in set (0.006 sec)
```

5)


```
mysql> SELECT CourseNo,
-> MAX(Score) AS MaximumMark,
-> MIN(Score) AS MinimumMark,
-> AVG(Score) AS AverageMark
-> FROM StudentMarks
-> GROUP BY CourseNo;
```

CourseNo	MaximumMark	MinimumMark	AverageMark
C001	75	55	65.0000
C002	82	48	69.2000
C003	88	67	77.5000
C004	90	90	90.0000

```
4 rows in set (0.022 sec)
```

6)

```
mysql> SELECT Course.CourseName,
-> AVG(StudentMarks.Score) AS AverageMark
-> FROM Course
-> JOIN StudentMarks
-> ON Course.CourseID = StudentMarks.CourseNo
-> GROUP BY Course.CourseName;
```

CourseName	AverageMark
Statistics	65.0000
Programming	69.2000
DBMS	77.5000
AI	90.0000

```
4 rows in set (0.008 sec)
```

7)

```
mysql> SELECT SUM(Score) AS TotalScore
-> FROM StudentMarks;
```

TotalScore
1442

```
1 row in set (0.006 sec)
```

8)

```
mysql> SELECT Course.CourseName,
-> COUNT(StudentMarks.RegNo) AS TotalStudents
-> FROM Course
-> JOIN StudentMarks
-> ON Course.CourseID = StudentMarks.CourseNo
-> GROUP BY Course.CourseName;
```

CourseName	TotalStudents
Statistics	4
Programming	10
DBMS	4
AI	2

4 rows in set (0.012 sec)

9)

```
mysql> SELECT StudentName,
-> COUNT(CourseNo) AS TotalCourses
-> FROM StudentMarks
-> GROUP BY StudentName;
```

StudentName	TotalCourses
Rahul	4
Priya	4
Arun	4
Anu	2
Kiran	2
Ram	2
Meena	2

7 rows in set (0.006 sec)

Ex. No.: 7

SELECT with various clause – GROUP BY, HAVING, ORDER BY

1)

```
mysql> SELECT Course.CourseName,
-> COUNT(Student.RegNo) AS TotalStudents
-> FROM Course
-> JOIN Student
-> ON Course.CourseID = Student.CourseID
-> GROUP BY Course.CourseName;
```

CourseName	TotalStudents
Statistics	3
Programming	2
DBMS	2

3 rows in set (0.006 sec)

2)

```
mysql> SELECT RegNo,
-> Name,
-> COUNT(CourseID) AS TotalCourses
-> FROM Student
-> GROUP BY RegNo, Name;
```

RegNo	Name	TotalCourses
191711341	Ramesh	1
191711342	Harish	1
191711343	Aish	1
191711344	Priya	1
191711345	Shyam	1
191711346	Kamal	1
191711347	Ram	1

7 rows in set (0.011 sec)

3)

```
mysql> SELECT *
-> FROM Faculty
-> ORDER BY Age DESC;
```

FacultyID	Name	DOJ	Age
101	Mohan	2012-05-20	59
102	Anitha	2016-08-15	45
103	Ravi	2014-10-10	NULL
104	Kumar	2013-12-01	NULL

4 rows in set (0.005 sec)

4)

```
mysql> SELECT StudentName, Score  
-> FROM StudentMarks  
-> WHERE Score >  
-> (  
-> SELECT AVG(Score)  
-> FROM StudentMarks  
-> );
```

StudentName	Score
Rahul	75
Rahul	82
Priya	88
Anu	90
Meena	79
Rahul	75
Rahul	82
Priya	88
Anu	90
Meena	79

```
10 rows in set (0.051 sec)
```